

Lab Assignment 2

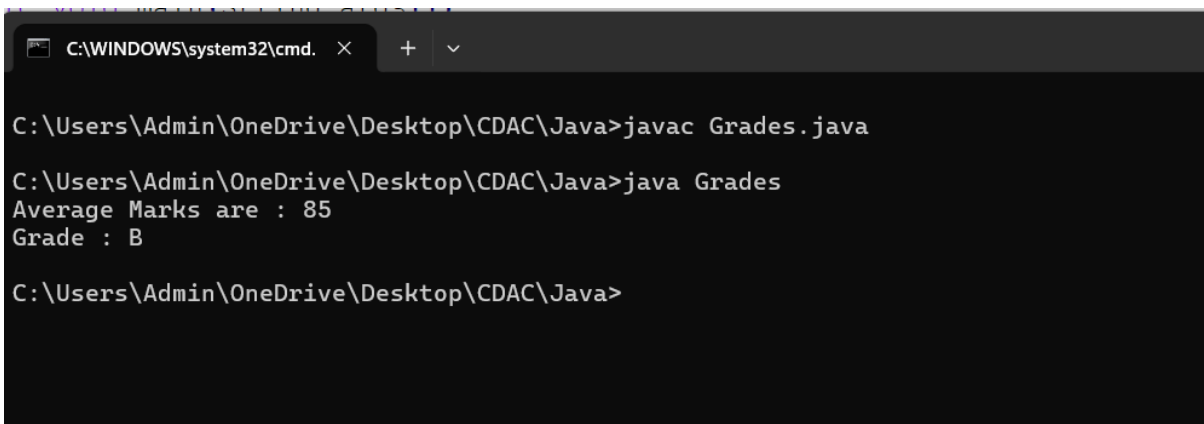
1. Problem 1: Grade Evaluation System

1.1 Code of Grade Evaluation System

```
public class Grades
{
    public static void main(String args[])
    {
        int maths = 80, science = 85, history = 90;
        int avg = (maths + science + history) / 3;
        System.out.println("Average Marks are : " + avg);

        if(avg >= 90)
            System.out.println("Grade : A");
        else if(avg >= 70 && avg <= 89)
            System.out.println("Grade : B");
        else if(avg >= 50 && avg <= 69)
            System.out.println("Grade : C");
        else if(avg >= 30 && avg <= 49)
            System.out.println("Grade : D");
        else
            System.out.println("Fail");
    }
}
```

1.2 Output of Grade Evaluation System



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac Grades.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java Grades
Average Marks are : 85
Grade : B

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Fig. 1.2- Output of Grade Evaluation System

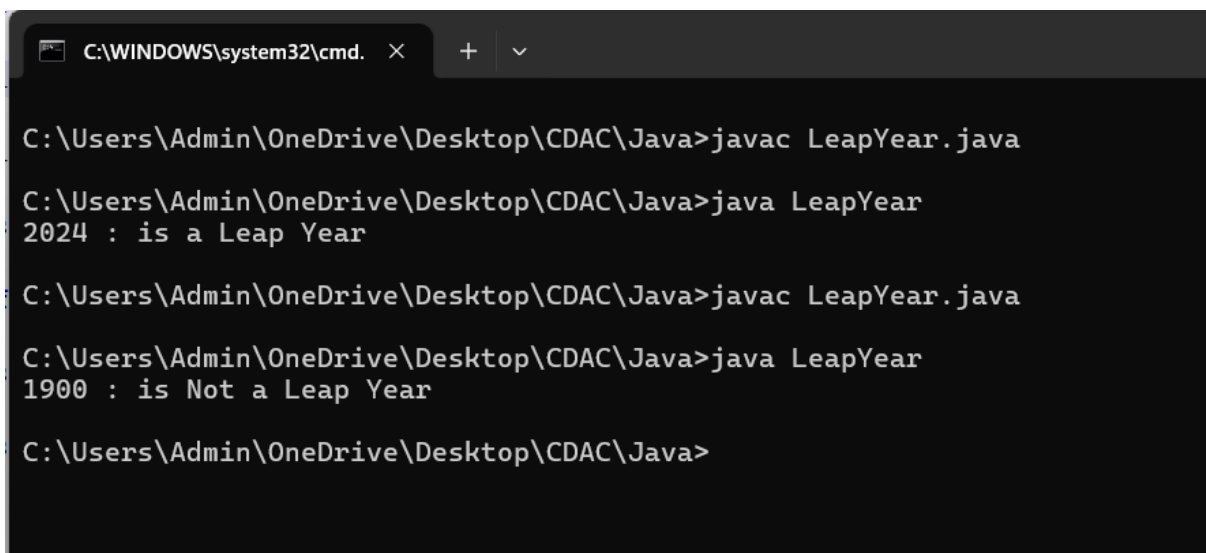
2. Problem 2: Leap Year

2.1 Code of Leap year

```
public class LeapYear
{
    public static void main(String args[])
    {
        int year = 2024;

        if( year % 400 == 0)
        {
            System.out.println(year + " : is a Leap Year");
        }
        else if(year % 4 == 0 && year % 100 != 0)
        {
            System.out.println(year + " : is a Leap Year");
        } else {
            System.out.println(year + " : is Not a Leap Year");
        }
    }
}
```

2.2 Output of Leap Year



```
C:\WINDOWS\system32\cmd. x + v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac LeapYear.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java LeapYear
2024 : is a Leap Year

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac LeapYear.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java LeapYear
1900 : is Not a Leap Year

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

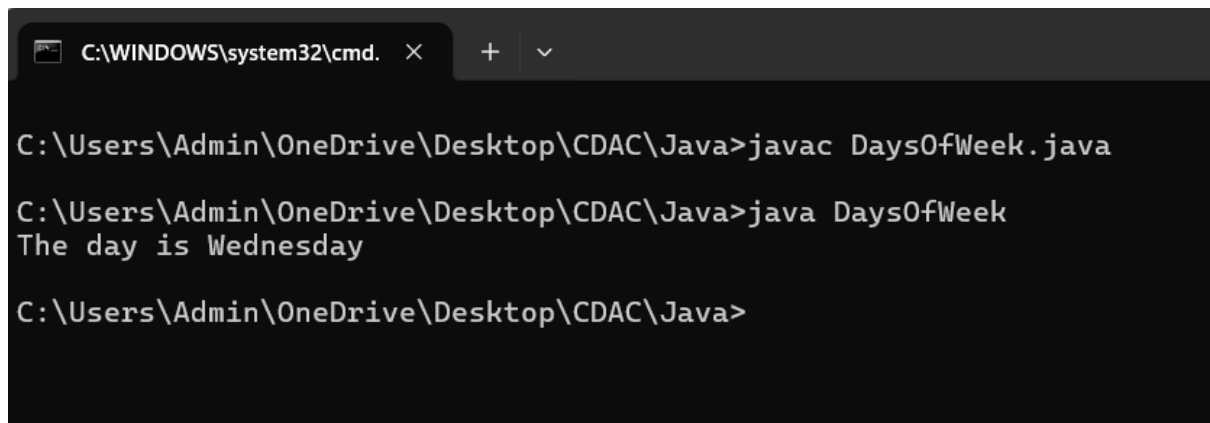
Fig. 2.2 Output of Leap Year

3. Problem 3: Days of the Week

3.1 Code of Days of the Week

```
public class DaysOfWeek
{
    public static void main(String args[])
    {
        int day = 3;
        switch(day)
        {
            case 1:
                System.out.println("The day is Monday");
                break;
            case 2:
                System.out.println("The day is Tuesday");
                break;
            case 3:
                System.out.println("The day is Wednesday");
                break;
            case 4:
                System.out.println("The day is Thursday");
                break;
            case 5:
                System.out.println("The day is Friday");
                break;
            case 6:
                System.out.println("The day is Saturday");
                break;
            case 7:
                System.out.println("The day is Saturday");
            default:
                System.out.println("Invalid day number");
        }
    }
}
```

3.2 Output of Days of the Week



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac DaysOfWeek.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java DaysOfWeek
The day is Wednesday

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Fig.3.2 Output of Days of the Week

4. Problem 4: Identify the Values of Uninitialized Variables

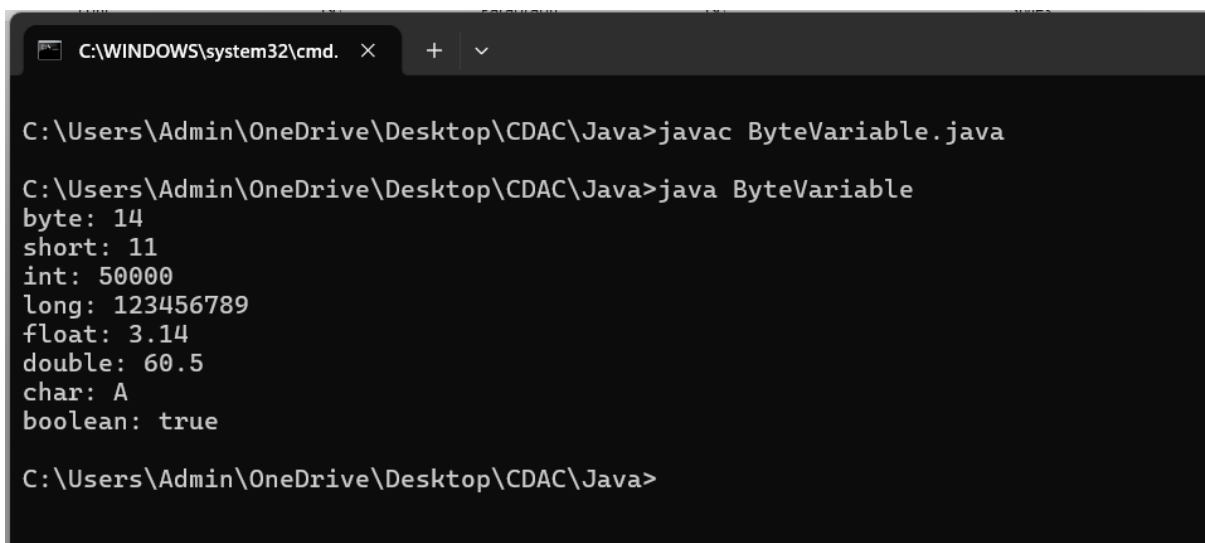
4.1 Code of Identify the Values of Uninitialized Variables

```
public class ByteVariable
{
    public static void main(String[] args)
    {

        byte a = 14;
        short b = 11;
        int c = 50000;
        long d = 123456789L;
        float e = 3.14f;
        double f = 60.5;
        char g = 'A';
        boolean h = true;

        System.out.println("byte: " + a);
        System.out.println("short: " + b);
        System.out.println("int: " + c);
        System.out.println("long: " + d);
        System.out.println("float: " + e);
        System.out.println("double: " + f);
        System.out.println("char: " + g);
        System.out.println("boolean: " + h);
    }
}
```

4.2 Out of Identify the Values of Uninitialized Variables



```
C:\WINDOWS\system32\cmd. X + v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac ByteVariable.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java ByteVariable
byte: 14
short: 11
int: 50000
long: 123456789
float: 3.14
double: 60.5
char: A
boolean: true

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

Fig.4.2 Out of Identify the Values of Uninitialized Variables